LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Fifth laboratory (Variant 5.1)

SPI programming of the 32 bit microcontrollers

Mandatory part (6 points):

Transfer the 4 byte of data using the SPI interface by performing the following steps:

- Configure the SPI4 interface: Full Duplex Master mode; Hardware NSS Signal – disable. Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity – Low; Clock Phase – 1 Edge; NSS Signal Type – Software. (1 point)
- Create the necessary variables for data transmission and reception. Values to transfer: 8, 12, 13, 15. (**1 point**)
- Connect the required pins with wires for data transmission from SPI4_MOSI (PE6) to SPI4_MISO (PE5). We will send and receive data to the same SPI4. SPI4_SCK pin is PE2. (1 point)
- Use the *HAL_SPI_TransmitReceive* function, which works in CPU blocking/pooling mode in order to perform the transmission. (**2 points**)
- Use the Debug configuration mode to demonstrate the values of the variables in the Live Expressions window and additionally show the transmitted and received bytes on the TFT screen. (1 point)

Additional part (4 points):

- Make the correct configuration of the project in order to use SPI4 and SPI3 communication lines. The following parameters should be selected during the configuration (1 point):

  - ✓ SPI4 should work in – Full Duplex Master mode; Hardware NSS Signal – disabled; Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity – Low; Clock Phase – 1 Edge; NSS Signal Type – Software.

- ✓ SPI3 should work in – Receive Only in Slave mode; Hardware NSS Signal – disabled; Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity – Low; Clock Phase – 1 Edge; NSS Signal Type – Software. NVIC Settings -> SPI3 Global Interrupt – enable.
- ✓ Configure TIM2 to capture time intervals of 1 second. Select the system clock and other required parameters so that the timer generates a 1 s period.

- Connect the required SPI4 and SPI3 pins with wires. Because SPI3 works in Receive Only Slave mode, it is sufficient to connect SPI4_SCK (PE2) to SPI3_SCK (PB3) and SPI4_MOSI (PE6) to SPI3_MOSI (PC12). For a symmetric test setup you may additionally connect SPI4_MISO (PE5) to SPI3_MISO (PC11). (1 point)
- Write a program in order to send 1 byte of data from SPI4 to SPI3 every 1 second from the timer/counter interrupt function. SPI3 should use an interrupt mechanism, in other words it should work in CPU Non Blocking Mode. Values 1, 3, 7 and 12 must be sent periodically from SPI4. The received results on SPI3 should be displayed on the TFT screen, while LEDs PG13 and PG14 are used as transfer activity indicators (2 points).

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Fifth laboratory (Variant 5.2)

SPI programming of the 32 bit microcontrollers

Mandatory part (6 points):

Transfer the 4 byte of data using the SPI interface by performing the following steps:

- Configure the SPI3 interface: Full Duplex Master mode; Hardware NSS Signal – disable. Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity – Low; Clock Phase – 1 Edge; NSS Signal Type – Software. (1 point)
- Create the necessary variables for data transmission and reception. Values to transfer: 1, 3, 11, 15. (**1 point**)
- Connect the required pins with wires for data transmission from SPI3_MOSI (PC12) to SPI3_MISO (PC11). We will send and receive data to the same SPI3. SPI3_SCK pin is PB3. (1 point)
- Use the *HAL_SPI_TransmitReceive* function, which works in CPU blocking/pooling mode in order to perform the transmission. (**2 points**)
- Use the Debug configuration mode to demonstrate the values of the variables in the Live Expressions window and additionally show the transmitted and received bytes on the TFT screen. (1 point)

Additional part (4 points):

- Make the correct configuration of the project in order to use SPI3 and SPI4 communication lines. The following parameters should be selected during the configuration (1 point):

  - ✓ SPI3 should work in – Full Duplex Master mode; Hardware NSS Signal – disabled; Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity – Low; Clock Phase – 1 Edge; NSS Signal Type – Software.

- ✓ SPI4 should work in – Receive Only in Slave mode; Hardware NSS Signal – disabled; Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity – Low; Clock Phase – 1 Edge; NSS Signal Type – Software. NVIC Settings -> SPI4 Global Interrupt – enable.
- ✓ Configure TIM3 to capture time intervals of 1 second. Select the system clock and other required parameters so that the timer generates a 1 s period.

- Connect the required SPI3 and SPI4 pins with wires. Because SPI4 works in Receive Only Slave mode, it is sufficient to connect SPI3_SCK (PB3) to SPI4_SCK (PE2) and SPI3_MOSI (PC12) to SPI4_MOSI (PE6). For a symmetric test setup you may additionally connect SPI3_MISO (PC11) to SPI4_MISO (PE5). (1 point)
- Write a program in order to send 1 byte of data from SPI3 to SPI4 every 1 second from the timer/counter interrupt function. SPI4 should use an interrupt mechanism, in other words it should work in CPU Non Blocking Mode. Values 15, 14, 13 and 12 must be sent periodically from SPI3. The received results on SPI4 should be displayed on the TFT screen, while LEDs PG13 and PG14 are used as transfer activity indicators (2 points).

Fifth laboratory (Variant 5.3)

SPI programming of the 32 bit microcontrollers

Mandatory part (6 points):

Transfer the 4 byte of data using the SPI interface by performing the following steps:

- Configure the SPI4 interface: Full Duplex Master mode; Hardware NSS Signal – disable. Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity – Low; Clock Phase – 1 Edge; NSS Signal Type – Software. (1 point)
- Create the necessary variables for data transmission and reception. Values to transfer: 4, 5, 7, 12. (**1 point**)
- Connect the required pins with wires for data transmission from SPI4_MOSI (PE6) to SPI4_MISO (PE5). We will send and receive data to the same SPI4. SPI4_SCK pin is PE2. (1 point)
- Use the *HAL_SPI_TransmitReceive* function, which works in CPU blocking/pooling mode in order to perform the transmission. (**2 points**)
- Use the Debug configuration mode to demonstrate the values of the variables in the Live Expressions window and additionally show the transmitted and received bytes on the TFT screen. (1 point)

Additional part (4 points):

- Make the correct configuration of the project in order to use SPI4 and SPI3 communication lines. The following parameters should be selected during the configuration (1 point):

  - ✓ SPI4 should work in – Full Duplex Master mode; Hardware NSS Signal – disabled; Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity – Low; Clock Phase – 1 Edge; NSS Signal Type – Software.
  - ✓ SPI3 should work in – Receive Only in Slave mode; Hardware NSS Signal – disabled; Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity

– Low; Clock Phase – 1 Edge; NSS Signal Type – Software. NVIC Settings -> SPI3 Global Interrupt – enable.

- ✓ Configure TIM2 to capture time intervals of 1 second. Select the system clock and other required parameters so that the timer generates a 1 s period.

- Connect the required SPI4 and SPI3 pins with wires. Because SPI3 works in Receive Only Slave mode, it is sufficient to connect SPI4_SCK (PE2) to SPI3_SCK (PB3) and SPI4_MOSI (PE6) to SPI3_MOSI (PC12). For a symmetric test setup you may additionally connect SPI4_MISO (PE5) to SPI3_MISO (PC11). (1 point)
- Write a program in order to send 1 byte of data from SPI4 to SPI3 every 1 second from the timer/counter interrupt function. SPI3 should use an interrupt mechanism, in other words it should work in CPU Non Blocking Mode. Values 1, 2, 3 and 4 must be sent periodically from SPI4. The received results on SPI3 should be displayed on the TFT screen, while LEDs PG13 and PG14 are used as transfer activity indicators (2 points).

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Fifth laboratory (Variant 5.4)

SPI programming of the 32 bit microcontrollers

Mandatory part (6 points):

Transfer the 4 byte of data using the SPI interface by performing the following steps:

- Configure the SPI3 interface: Full Duplex Master mode; Hardware NSS Signal – disable. Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity – Low; Clock Phase – 1 Edge; NSS Signal Type – Software. (1 point)
- Create the necessary variables for data transmission and reception. Values to transfer: 7, 8, 10, 11. (**1 point**)
- Connect the required pins with wires for data transmission from SPI3_MOSI (PC12) to SPI3_MISO (PC11). We will send and receive data to the same SPI3. SPI3_SCK pin is PB3. (1 point)
- Use the *HAL_SPI_TransmitReceive* function, which works in CPU blocking/pooling mode in order to perform the transmission. (**2 points**)
- Use the Debug configuration mode to demonstrate the values of the variables in the Live Expressions window and additionally show the transmitted and received bytes on the TFT screen. (1 point)

Additional part (4 points):

- Make the correct configuration of the project in order to use SPI3 and SPI4 communication lines. The following parameters should be selected during the configuration (1 point):

  - ✓ SPI3 should work in – Full Duplex Master mode; Hardware NSS Signal – disabled; Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity – Low; Clock Phase – 1 Edge; NSS Signal Type – Software.
  - ✓ SPI4 should work in – Receive Only in Slave mode; Hardware NSS Signal – disabled; Frame Format – Motorola; Data Size – 8 bits; Prescaler (for Baud rate) – 8; Clock Polarity

– Low; Clock Phase – 1 Edge; NSS Signal Type – Software. NVIC Settings -> SPI4 Global Interrupt – enable.

- ✓ Configure TIM3 to capture time intervals of 1 second. Select the system clock and other required parameters so that the timer generates a 1 s period.

- Connect the required SPI3 and SPI4 pins with wires. Because SPI4 works in Receive Only Slave mode, it is sufficient to connect SPI3_SCK (PB3) to SPI4_SCK (PE2) and SPI3_MOSI (PC12) to SPI4_MOSI (PE6). For a symmetric test setup you may additionally connect SPI3_MISO (PC11) to SPI4_MISO (PE5). (1 point)
- Write a program in order to send 1 byte of data from SPI3 to SPI4 every 1 second from the timer/counter interrupt function. SPI4 should use an interrupt mechanism, in other words it should work in CPU Non Blocking Mode. Values 15, 7 and 3 must be sent periodically from SPI3. The received results on SPI4 should be displayed on the TFT screen, while LEDs PG13 and PG14 are used as transfer activity indicators (2 points).